P.E.S. COLLEGE OF ENGINEERING,

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MANDYA-571401

MINI PROJECT REPORT ON

"Analysis of Sentiment Detection Tool"

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Abstract

This project presents a machine-learning-based **Sentiment Analysis Tool** designed to classify user-generated text as *positive, negative,* or *neutral*. With the rapid growth of social media, customer reviews and online feedback systems, understanding public opinion automatically has become essential for businesses, organizations and research communities. Manual analysis of large-scale text data is time-consuming, inaccurate and highly subjective.

The proposed system uses pre-trained transformer models such as **BERT** and **RoBERTa** to analyse sentence-level sentiments with high accuracy. A lightweight web interface is implemented, allowing users to enter text, submit it for analysis and instantly view the sentiment classification. Additionally, the tool supports **visual sentiment analytics**, where trends are plotted in the form of charts or graphs. This system provides efficient, real-time sentiment insights and can be adapted for tweets, reviews or custom datasets.

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Introduction

Sentiment analysis, often called opinion mining, is a widely used AI technique for determining the emotional tone behind textual data. With the explosion of social media platforms, people openly express their opinions about products, movies, services and public events. Analysing this data manually becomes unrealistic when millions of posts are generated every minute.

Traditional machine learning techniques like Naive Bayes and SVM have shown moderate performance but fail to capture contextual meaning, sarcasm and linguistic variations. Modern transformer-based models such as **BERT**, **RoBERTa** and **DistilBERT** provide state-of-the-art text understanding, enabling more accurate sentiment classification.

The goal of this project is to design a simple, scalable and user-friendly **web-based sentiment analysis tool** that processes user input, classifies the sentiment and displays outputs along with visual graphs. The tool can be used for social media monitoring, customer feedback evaluation, product review analysis and academic research involving opinion studies

Literature Survey

Sentiment analysis has been extensively studied due to its importance in natural language processing (NLP) applications. Early research by Pang & Lee (2002) introduced machine-learning techniques for movie review polarity detection, demonstrating that SVMs outperform traditional rule-based methods. Liu B. (2012) explored opinion mining and sentiment classification techniques, highlighting the challenges in handling ambiguity and subjectivity.

The introduction of transformer models revolutionized NLP. Devlin et al. (2019) presented **BERT (Bidirectional Encoder Representations from Transformers)**, which significantly improved sentiment classification accuracy through contextual embeddings. Later, Liu et al. (2019) proposed **RoBERTa**, an optimized training approach achieving better performance on several benchmark sentiment datasets.

Recent advancements include sentiment visualization tools and dashboards for customer behaviour analysis. Studies also show that domain-adapted models outperform generic sentiment classifiers when applied to specialized categories such as medical reviews or financial sentiment. The proposed project incorporates these modern techniques by using pre-trained transformer models and a web-based interactive interface for real-time sentiment evaluation.

Problem Statement

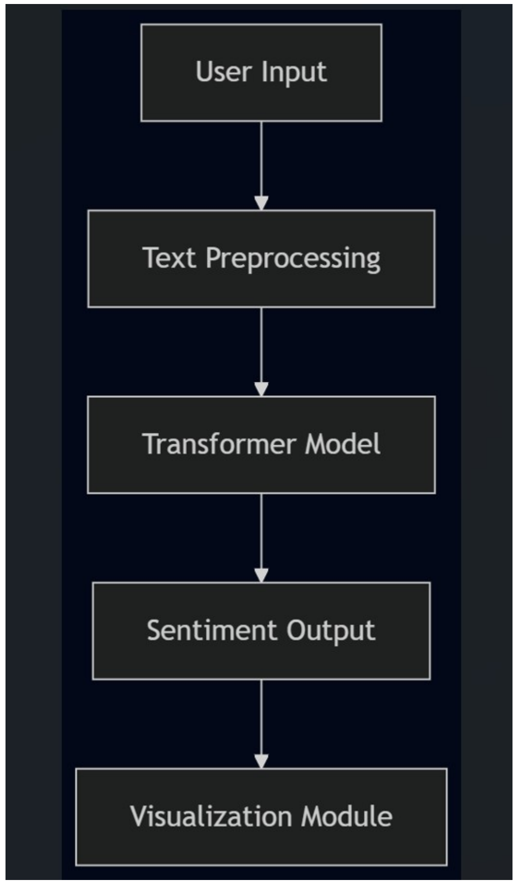
With the rapid growth of digital communication, users frequently post opinions, reviews and feedback online. However, organizations and researchers struggle to manually process large volumes of text data to understand public sentiment. Traditional sentiment analysis models lack contextual understanding and often produce inaccurate results when exposed to slang, sarcasm or complex sentence structures.

There is a need for a **modern, AI-powered sentiment analysis tool** capable of understanding context, providing accurate sentiment classification and visualizing trends over time. Existing tools are either domain-specific, require technical expertise, or charge subscription fees. Therefore, a customizable, user-friendly sentiment analysis system is required to analyze text from tweets, reviews and general user input in real-time.

Objectives

1. **To study and analyze** existing sentiment analysis techniques, including traditional machine learning approaches and modern transformer-based models such as **BERT** and **RoBERTa**.
2. **To implement a sentiment analysis system** using pre-trained transformer models capable of classifying text into **positive, negative, or neutral** categories with high accuracy.
3. **To evaluate the performance** of the implemented models using standard classification metrics such as **accuracy, precision, recall, and F1-score**, and identify the most suitable model for real-time sentiment analysis.
4. **To develop a user-friendly web interface** that allows users to input text and receive instant sentiment predictions along with confidence scores.
5. **To incorporate data visualization** to represent sentiment trends and distributions using graphical formats such as **bar charts and pie charts** for better interpretability.
6. **To test the system** on diverse datasets including social media posts, product reviews, and custom user inputs to ensure consistency and robustness across different text domains.

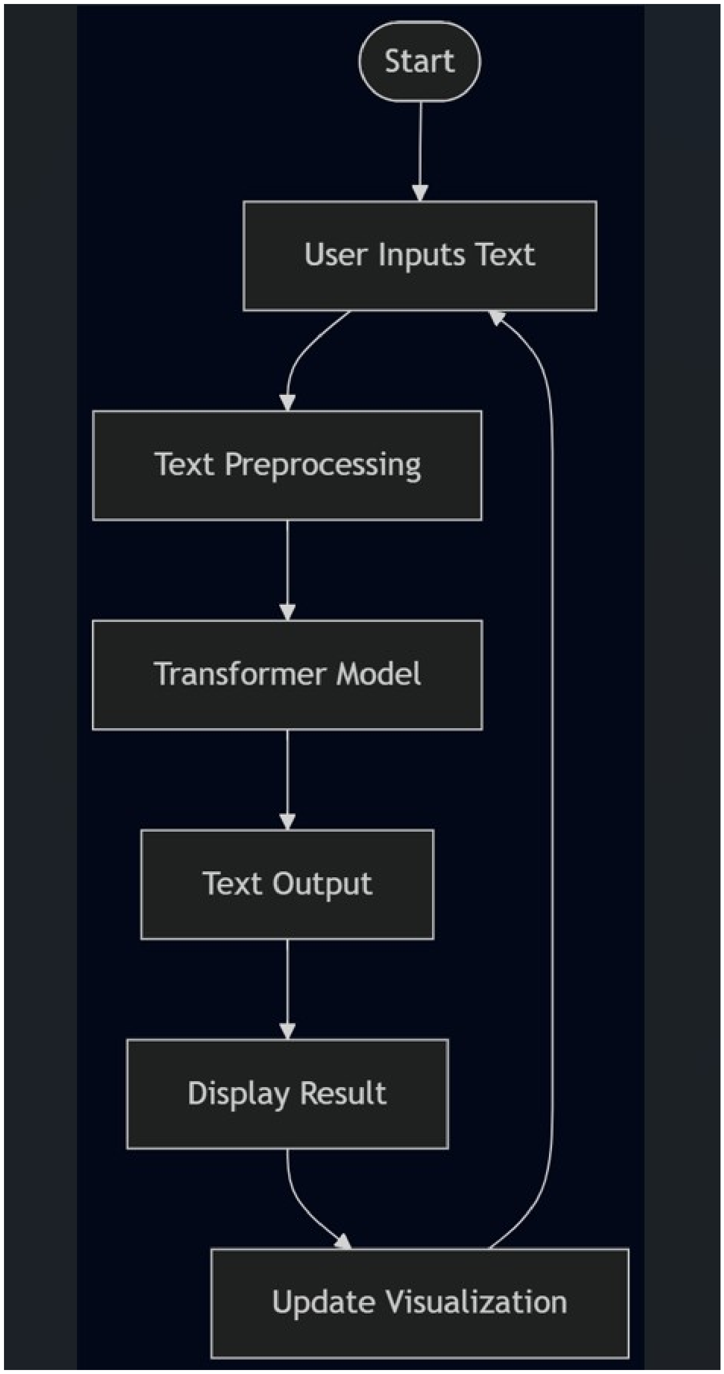
System Block Diagram / Methodology



Workflow:

1. User enters text (review, tweet or sentence) into the web interface.
2. The text is pre-processed: tokenization, cleaning and normalization.
3. A pre-trained transformer model (BERT/RoBERTa) predicts the sentiment.
4. Backend processes the output and returns a sentiment label (positive/negative/neutral).
5. Sentiment results are stored temporarily for visualization.
6. Charts/graphs display sentiment distribution and trends to the user.

Flow Chart with Working Mechanism



Expected Outcome

- A fully functional **real-time sentiment analysis tool**.
- Accurate sentiment predictions powered by transformer-based AI models.
- A simple and responsive web interface for sentiment input.
- Visualization of sentiment insights using line charts, pie charts or bar graphs.
- Ability to analyse tweets, reviews or custom user inputs.
- Extendable framework for future domain-specific sentiment analysis applications.

References

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- **Y. Liu et al., "RoBERTa:** A Robustly Optimized BERT Pretraining Approach," *arXiv preprint arXiv:1907.11692*, 2019.

Feedback Form

- The synopsis is formatted as per the template. Yes/No.
- Objectives and problem statement are properly formulated. Yes/No.
- Picture-diagrams-tables are numbered properly. Yes/No.
- Syntax and Grammar usage is standard. Yes/No.
- References are cited properly and listed in order. Yes/No.
- Remarks and comments.

Signature of the Guide **Signature of the Mini-Project Coordinator**